For parents and teachers — child activity page for adult-led use

Name: _____

Which Falls Faster: Flat or Crumpled Paper?



1 My guess

- ☐ Flat sheet first
- ☐ Crumpled ball first
- ☐ Same time

I think so because _____

_____

2 Fair test

Use matching paper. Keep the height and release time the same. **Change only the shape.** Use two matching sheets; crumple one.



3 Try it three times

Try	Which landed first? (flat / ball / same time)
1	
2	
3	

4 What did we find?

The _____ landed first in _____ out of 3 tries.

5 Think further

What might happen on the Moon, where there is almost no air?

_____

_____

How could a parachute slow a fall?

_____

_____

SAFETY Ask an adult to drop both papers while standing on the floor. Never climb.

Guess Breakers · Guess. Test. Find out!

Watch the episode first · youtube.com/watch?v=8q4nmNSRnk8

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Adult Guide



Goal and time: About 15 minutes. Children make a prediction, run a fair comparison, and explain what the evidence shows.

Connects to the investigation practice in NGSS 3-PS2-1; this activity does not assess the full standard.

Materials: Two matching sheets of paper; a clear floor-level drop space.

How to lead it

1. Watch the episode together. Ask each child to make a guess and explain why.
2. Keep one sheet flat. Crumple the matching sheet into a ball.
3. Work indoors, away from drafts and fans. One adult holds one paper in each hand, side by side at the same height. Hold the flat sheet horizontally. Release both as nearly together as possible, without pushing. If one is clearly released first, repeat that try.
4. Repeat three times and record what landed first. Discuss what stayed the same and what changed.

> **Make it fair:** Use matching sheets of the same size and type. Keep the drop height and release time the same. Change only the shape. Let go together without pushing. Real drops can vary; repeat and compare what happens.

Expected result and why

The crumpled ball lands first, while the flat sheet falls more slowly. Crumpling does not make the paper heavier: the sheets have the same mass. Air pushes against falling paper. The flat sheet meets more air, so air resistance slows it more. On the Moon, where there is almost no air, Apollo 15 astronaut David Scott dropped a hammer and a feather; they landed together within the demonstration's precision.

The paper drop in the episode is an animated demonstration of a classroom experiment, not new measurements. Its Apollo 15 scene is a stylized animated reconstruction, not real footage. The paper trial is a model for thinking about air resistance; it is not a Moon experiment.

Discuss

- What did we keep the same? What did we change?
- Did every try match your guess? What does the evidence tell you?
- Why can a wide parachute slow a fall?

Sources

University of Wisconsin–Madison, Ball, Paper and Cotton-Ball Drop
https://wiki.physics.wisc.edu/facultywiki/BallPaperCottonBallDrop

University of Colorado Boulder, Book and Paper
https://physicslabs.colorado.edu/demos/mechanics/motion-in-one-dimension/uniform-acceleration/book-and-paper/

NASA NSSDC, Apollo 15 Hammer-Feather Drop
https://nssdc.gsfc.nasa.gov/planetary/lunar/apollo_15_feather_drop.html

NASA, The Moon's Atmosphere
https://science.nasa.gov/moon/lunar-atmosphere/

NASA, Orion Parachutes
https://www.nasa.gov/podcasts/houston-we-have-a-podcast/orion-parachutes/

AI note: The episode is animated, and its voices are synthesized with AI. Animation and synthetic voices do not require an AI label under the channel's policy.

No child data, submissions, or contact details are requested.

Free to print for classroom and home use.

Guess Breakers
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